

Practice Problems (Loop)

1. Print All Prime Numbers Within a Given Range [a, b]

Write a program to input two integers a and b and print all prime numbers between them. A number is prime if it is greater than 1 and has no divisors other than 1 and itself. Use a loop to check divisibility for each number.

2. Generate and Display the Fibonacci Sequence Up to the n th Term

The Fibonacci sequence starts with 0 and 1, and each subsequent term is the sum of the previous two terms. Input n and print the first n terms of the Fibonacci sequence. Use a loop to calculate each term iteratively.

3. Check if a Given Number Is an Armstrong Number

An Armstrong number is a number that is equal to the sum of its digits raised to the power of the number of digits. For example, $153 = 1^3 + 5^3 + 3^3$. Input a number, calculate the sum of its digit powers, and check if it equals the original number.

4. Reverse the Digits of a Given Integer and Print the Result

Input a number and reverse its digits using a loop. For example, if the input is 1234, the output should be 4321. Use modulo (%) and integer division (/) to extract and reverse digits.

5. Compute the Sum of All Odd Numbers Between 1 and a Given Number n

Input an integer n and calculate the sum of all odd numbers between 1 and n . Use a loop to iterate through numbers and add them to the sum if they are odd.

6. Count the Number of Digits in a Given Integer

Input a number and count how many digits it has. Use a loop to divide the number by 10 until it becomes 0, keeping a counter for each iteration.

7. Print the First n Terms of a Series Where Each Term Is Defined as $S_n = n^2 + 1$

Input an integer n and print the first n terms of the series $S_n = n^2 + 1$. For example, if $n = 4$, the output should be 2, 5, 10, 17.

8. Find the Greatest Common Divisor (GCD) of Two Integers Using a Loop

Input two integers x and y and calculate their GCD using a loop. GCD is the largest number that divides both x and y without leaving a remainder. Use the Euclidean algorithm iteratively.

9. Print the Factorial of a Number Without Using Recursion

Input a number and calculate its factorial using a loop. The factorial of n is the product of all integers from 1 to n . For example, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

10. Determine If a Number Is a Palindrome

Input a number and check if it reads the same backward as forward. Reverse the number using a loop (as in Problem 4) and compare it to the original number.

11. Print the Sum of the Series $1 - 2 + 3 - 4 \dots$ upto n th term

Input an integer n and calculate the sum of the above series up to the n th term.

12. Print All Numbers Divisible by 7 but Not Divisible by 5 Within a Range [a, b]

Input two integers a and b and print all numbers in the range that are divisible by 7 but not divisible by 5. Use a loop to iterate through the range and apply the conditions.

13. Find the Largest Digit in a Given Integer

Input a number and find its largest digit using a loop. For example, for 4573, the largest digit is 7. Extract digits using modulo and compare them to track the largest.

14. Print the Sum of the Series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$

Input an integer n and calculate the sum of the harmonic series up to the n th term. Use a loop to add $\frac{1}{i}$ for each i from 1 to n .

15. Display the Sum of All Numbers Less Than n That Are Multiples of 3 or 5

Input an integer n and calculate the sum of all numbers less than n that are divisible by 3 or 5. For example, if $n = 10$, the numbers are 3, 5, 6, 9, and their sum is 23.